

Samba server types and security modes

There are three different types of servers available:

- **Domain controller**
 - Primary domain controller (PDC)
 - Backup domain controller (BDC)
 - ADS domain controller
- **Domain member server**
 - Active directory domain server
 - NT4 style domain server
- **Standalone server**

There are two types of security modes

available for Samba: share-level and user-level, known as security levels.

Share-level security: This can be implemented in only one way. With share-level security, the server expects a password for each share. It is not dependent on a particular user name.

In *smb.conf*, the security=share directive that sets share-level security is:

```
[GLOBAL]
security = share
```

User-level security: This is the default setting for Samba. It is used by Samba, even if the security = user directive is not listed in the *smb.conf* file. Once the server accepts the user name/password, the user can then access multiple shares without entering the password for each instance.

In *smb.conf*, the security = user directive that sets user-level security is:

```
[GLOBAL]
security = user
```

There are different implementations of user level security (see Figure 3).

Domain security mode: In this mode, the Samba server has a domain security trust account, which ensures all authentication requests are handled by domain controllers.

It can be achieved by using the following directives in *smb.conf*:

```
[GLOBAL]
...
security = domain
workgroup = MARKETING
```

Active directory security mode: In the active directory environment, this mode is used to join as a native active directory member. The Samba server can join an ADS using Kerberos, even if a security policy restricts the use of NT-

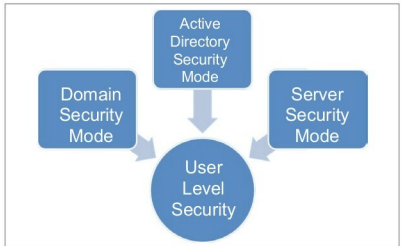


Figure 3: User level security modes

compatible authentication protocols.

In *smb.conf*, this can be achieved by using the following directives:

```
[GLOBAL]
...
security = ADS
realm = test.COM
password server = kerberos.test.com
```

Server security mode: This is used when Samba is not capable of acting as a domain member server. In *smb.conf*, this mode is achieved by using the following directives:

```
[GLOBAL]
...
encrypt passwords = Yes
security = server
password server = "pwd"
```



References

- [1] <https://www.samba.org/samba/>
- [2] https://docs.fedoraproject.org/en-US/Fedora/12/html/Deployment_Guide/ch-samba.html#samba-rgs-overview
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- [6] http://www.linuxtopia.org/online_books/network_administration_guides/samba_reference_guide/index.html

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